# Preferences and Utility Functions

## EC306: Intermediate Microeconomics — Day 1

Core reading: Perloff (2020), *Microeconomics: Theory and Applications with Calculus*, 5th ed., Ch. 3

# Where this day sits in the five-day arc

We spend five intensive days building the demand and supply curves from primitives.

1. Preferences → 2. Optimisation → 3. Demand → 4. Production → 5. Cost & supply

**Today's job:** describe what a consumer *wants* before we ask what they *can afford*. Get this layer wrong and every demand curve downstream is wrong.

# Roadmap for today

1. Preference relations and the three core axioms
2. Indifference curves: what their shape encodes
3. Marginal rate of substitution (MRS)
4. Utility as a *representation* — ordinal, not cardinal
5. Monotonic transformations
6. Five workhorse utility functions

### Today's running example

A student allocating a weekend between **streaming hours** ($x$) and **coffee** ($y$). We will keep returning to this to make each abstraction concrete.

# The primitive: a preference relation, not a number

We start from how a consumer *ranks* bundles, not from any number.
A bundle is $\mathbf{x} = (x_1, x_2)$. Write:

- $\mathbf{x} \succsim \mathbf{y}$ : "$\mathbf{x}$ is at least as good as $\mathbf{y}$"
- $\mathbf{x} \succ \mathbf{y}$ : strictly preferred
- $\mathbf{x} \sim \mathbf{y}$ : indifferent

## Why start here?

Numbers (utility) come *later*, and only as a convenient summary of the ranking. The economics lives in the ranking; the utility function is bookkeeping.

# The three axioms of “rational” preferences

### 1. Completeness

For any two bundles, either $\mathbf{x} \succsim \mathbf{y}$, or $\mathbf{y} \succsim \mathbf{x}$, or both. *The consumer can always compare.*

### 2. Transitivity

If $\mathbf{x} \succsim \mathbf{y}$ and $\mathbf{y} \succsim \mathbf{z}$, then $\mathbf{x} \succsim \mathbf{z}$. *No preference cycles.*

### 3. Continuity (technical)

Small changes in a bundle do not cause sudden jumps in ranking. This is what guarantees a *continuous* utility function exists.

# Transitivity is an assumption, not a law — and it can fail

**The money pump.** Suppose Maya's ranking cycles: espresso $\succ$ latte $\succ$ filter $\succ$ espresso.

- She has filter. You offer espresso for filter + £0.10. She accepts (espresso $\succ$ filter).
- You offer latte for espresso + £0.10. She accepts.
- You offer filter for latte + £0.10. She accepts …and is back to filter, 30p poorer.

## Real-world tension

Empirically, *intransitive* choices show up constantly — menu effects, framing, the "decoy" option that makes a mid-tier wine sell. The axioms are a *modelling* device, justified by predictive usefulness, not by being literally true of every human.

This is exactly the kind of caveat a chatbot omits and an examiner rewards. Keep it.

# Two more assumptions we usually add

## Monotonicity ("more is better")

If $\mathbf{x}$ has at least as much of every good and strictly more of one, then $\mathbf{x} \succ \mathbf{y}$.

*Consequence:* indifference curves slope **downward** and bundles to the north-east are preferred.

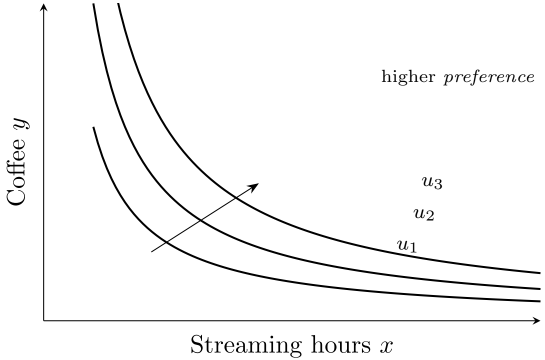
## Convexity ("averages beat extremes")

Mixtures are weakly preferred to extremes. If $\mathbf{x} \sim \mathbf{y}$, then any weighted average is at least as good.

*Consequence:* indifference curves are **bowed toward the origin**.

Monotonicity $\Rightarrow$ slope; convexity $\Rightarrow$ curvature. Together they pin down the familiar shape.

# Indifference curves: the geometry of the ranking




Each curve = a set of bundles among which the consumer is indifferent. Curves *never cross* (that would violate transitivity — ask why on the next slide).

# Why indifference curves cannot cross

Suppose two curves crossed at bundle $A$. Let $B$ sit on one curve, $C$ on the other.

- $A \sim B$ (same curve) and $A \sim C$ (same curve) $\Rightarrow$ by transitivity $B \sim C$.
- But $B$ lies strictly north-east of $C$, so monotonicity gives $B \succ C$.

Contradiction. So curves from a transitive, monotone preference cannot intersect.

This is a one-line proof you should be able to reproduce from scratch — it is a favourite exam "explain why" question.

# The slope has a name: the MRS

The **marginal rate of substitution** is the rate at which the consumer will trade $y$ for one more unit of $x$ while staying *indifferent*:

$$\text{MRS} \;=\; -\frac{\mathrm{d}y}{\mathrm{d}x}\bigg|_{u=\bar{u}}.$$

Along an indifference curve $u(x, y) = \bar{u}$, totally differentiate:

$$\frac{\partial u}{\partial x}\,\mathrm{d}x + \frac{\partial u}{\partial y}\,\mathrm{d}y = 0 \quad\Longrightarrow\quad \boxed{\text{MRS} = \frac{\partial u/\partial x}{\partial u/\partial y} = \frac{\text{MU}_x}{\text{MU}_y}}$$

## Read it in plain English

"How much coffee am I willing to give up for one more hour of streaming?" Diminishing MRS (convexity) says: the more streaming I already have, the less coffee I'll sacrifice for another hour.

# Worked example — MRS for Cobb–Douglas

Let $u(x, y) = x^{0.5}y^{0.5}$ (streaming and coffee).

$$\mathrm{MU}_x = \tfrac{1}{2}x^{-0.5}y^{0.5}, \qquad \mathrm{MU}_y = \tfrac{1}{2}x^{0.5}y^{-0.5}.$$

$$\mathrm{MRS} = \frac{\mathrm{MU}_x}{\mathrm{MU}_y} = \frac{\tfrac{1}{2}x^{-0.5}y^{0.5}}{\tfrac{1}{2}x^{0.5}y^{-0.5}} = \frac{y}{x}.$$

## Interpretation

At $(x, y) = (1, 4)$: $\mathrm{MRS} = 4$ — with little streaming, she'll give up 4 coffees for one more hour. At $(4, 1)$: $\mathrm{MRS} = 0.25$ — now only a quarter of a coffee. The trade-off *shrinks* as $x$ grows: diminishing MRS, exactly as convexity predicts.

# Your turn (1) — in-lecture

## Your turn

For $u(x, y) = x^2 y$:

1. Find $\mathrm{MU}_x$ and $\mathrm{MU}_y$.
2. Derive the MRS.
3. Evaluate it at $(2, 3)$ and state, in one sentence, what the number means.

# Utility is a *representation* of the ranking

A utility function $u(\cdot)$ **represents** a preference if

$$\mathbf{x} \succsim \mathbf{y} \quad \Longleftrightarrow \quad u(\mathbf{x}) \geq u(\mathbf{y}).$$

## Ordinal, not cardinal

Only the *order* of the numbers matters, not their size or spacing. "$u = 8$ vs $u = 4$" does **not** mean "twice as happy". That statement is meaningless in standard consumer theory.

**Consequence:** any strictly increasing transformation of $u$ represents the *same* preferences.

# Monotonic transformations leave behaviour unchanged

Take $u(x, y) = x^{0.5}y^{0.5}$ and transform it three ways:

$$v = u^2 = xy, \qquad w = \ln u = \tfrac{1}{2}\ln x + \tfrac{1}{2}\ln y, \qquad z = 2u + 7.$$

All four describe *identical* choices. Check the MRS is invariant:

$$\mathrm{MRS}_u = \frac{y}{x}, \qquad \mathrm{MRS}_v = \frac{y}{x}, \qquad \mathrm{MRS}_w = \frac{1/(2x)}{1/(2y)} = \frac{y}{x}.$$

## Why you care

ln of Cobb–Douglas turns products into sums — vastly easier to optimise tomorrow. We exploit this constantly. The transformation is free because behaviour is unchanged.

# Your turn (2) — in-lecture

## Your turn

A colleague claims: "Since $u_A = x + y$ gives bundle $(3, 3)$ a utility of 6 and $u_B = (x + y)^2$ gives it 36, person B likes the bundle more."

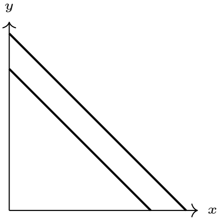
In two sentences, say what is wrong with this claim.

# Five functional forms you must recognise on sight

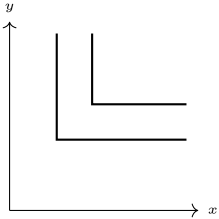
| Name | Form | Signature feature |
| --- | --- | --- |
| Cobb–Douglas | $x^a y^b$ | smooth convex; constant expenditure shares |
| Perfect substitutes | $ax + by$ | straight-line ICs; constant MRS $= a/b$ |
| Perfect complements | $\min\{ax, by\}$ | L-shaped; consumed in fixed ratio |
| Quasilinear | $v(x) + y$ | ICs are vertical shifts; no income effect on $x$ |
| CES | $(ax^\rho + by^\rho)^{1/\rho}$ | nests the above as $\rho$ varies |

Each shape encodes a different *substitutability* between goods.

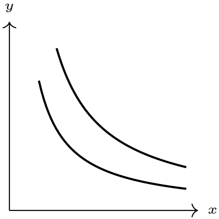
# The shapes, side by side




Substitutes




Complements




Cobb–Douglas

## Map to the real world

**Substitutes:** Pepsi vs Coke; petrol from two stations. **Complements:** left vs right shoes; printer vs cartridge. **Cobb–Douglas:** a household's split across broad categories (food, housing) — empirically shares are roughly stable, which is the Cobb–Douglas signature.

# Why “perfect complements” is the printer-cartridge model

$u = \min\{x, y\}$ for one printer ($x$) and one cartridge ($y$): a second cartridge with no printer adds nothing.

Optimal: $x = y$ always, regardless of prices.

## Pricing strategy fallout

This is *exactly* why firms give away printers and overprice cartridges (the “razor-and-blades” model). Because demand is locked to a fixed ratio, the firm shifts margin onto the good with weaker outside competition. The functional form predicts a real business model.

# Quasilinear: when one good has no income effect

$u(x, y) = v(x) + y$, e.g. $u = 4\sqrt{x} + y$, where $y$ is "money spent on everything else".

$$\text{MRS} = \frac{v'(x)}{1} = v'(x) \quad \text{depends on } x \text{ only.}$$

## Why it matters

Demand for $x$ is independent of income (above a threshold). This is the standard model for a "small" good — toothpaste, a coffee — where spending doesn't rise with your salary. It also makes welfare analysis clean, which is why it dominates applied policy work.

# Your turn (3) — in-lecture

## Your turn

Classify each pair and state the matching utility form:

1. Left shoes and right shoes
2. Two brands of bottled water a buyer regards as identical
3. Bus tickets and a monthly salary ("everything else")

# What you can now do

1. State the preference axioms and *argue* why each matters (and where it fails).
2. Prove indifference curves don't cross.
3. Derive and interpret the MRS for any differentiable utility.
4. Explain why utility is ordinal and use monotonic transforms freely.
5. Recognise five functional forms and the real markets they model.

**Next**

We add the **budget constraint** and ask: of all affordable bundles, which one does this consumer pick? That is the constrained optimisation problem — and it delivers the demand function.